Fiscal Year:	FY 2022	**Task Last Updated:**	FY 09/02/2021
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Project Title:	From the Perspective of the Target, a Cis-Regulatory Perspective on the Effects of Microgravity		
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Program/Discipline--Element/Subdiscipline:			
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Human Research Program Risks:	None		
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Comments:			
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No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA KSC
Contact Monitor:	Massa, Gioia	**Contact Phone:**	321-861-2938
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Key Personnel Changes/Previous PI:			
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Task Description:	Long duration spaceflight and future colonization missions will require plants to provide much needed resources for human health and well-being. Plants are amazingly hardy organisms which have over the years, developed mechanisms to perceive and respond to stresses they encounter. This evolution occurred in terrestrial environments here on Earth, and not in the environment of space. We are interested in understanding the mechanisms by which plants perceive and respond to stresses, such as a change in gravity or other aspects of spaceflight. Understanding how plants sense and respond to this environment is critical for enabling plant growth in space. We examined the available transcriptional data in GeneLab and observed that, while there were few genes that were cohesively expressed across all conditions, there was some consistency in the cis-regulatory features that are enriched. Based on findings from GeneLab Transcriptional experiments we propose to examine the effects of spaceflight through the lens of the cis-regulatory network. Our analysis of the Arabidopsis transcriptional datasets available in GeneLab shows that there is little overlap in the individual differentially expressed (DE) genes between the various spaceflight experiments. Rather than querying the overlap of specific genes across experiments, we are focusing on upstream signaling components, using the conserved cis-regulatory sequences as a tool to uncover the upstream regulators. We have identified enriched and conserved motifs in the promoters of genes differentially expressed in spaceflight experiments. This approach is inclusive and particularly beneficial for identifying patterns in limited and highly dissimilar data sets. We will take a focused approach, selecting a few of the motifs that are enriched in genes that are responsive to microgravity across multiple independent experiments in GeneLab and examine the cis-regulatory control from the bottom up. The objective of this study is to utilize the GeneLab repository as a starting point to identify mechanisms regulating plant responses to spaceflight and microgravity. Our specific aims are: 1. To screen a yeast-one-hybrid (Y1H) transcription factor library to identify specific transcription factors This objective is to screen the three motifs indicated in the proposal and simultaneously identify additional motifs, integrating additional data available from GeneLab, particularly those motifs enriched in the 5’ Untranslated Region (UTR) and first intron. 2. Confirm DNA-binding of transcription factors and obtain transcription factor mutants This objective is to test the identified candidate transcription factors that bind to the potential gravity responsive elements and confirm their binding to the promoter region in yeast and test in planta (objective 3) their effects on the induction of the reporters generated from these elements. 3. Generate promoter:reporter lines to test in planta In this objective we propose to develop reporter constructs for our top candidates in planta to test the response of these motifs. This proposal is in response to Appendix G of the Research Opportunities in Space Biology research announcement. Specifically, our proposal will address 4b) Systems Biology Experimental Research using the GeneLab Data System under "Investigators who wish to develop new experimental hypotheses based on data in the GeneLab database." We anticipate that this work will enable us to delineate signaling pathway/s responding to microgravity and yield new knowledge on plant specific responses to spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research provides a fundamental understanding of how the plant responds to environmental stimuli. Our cis-regulatory network and new approach developed in this proposal can be applied to stresses beyond gravity to identify the regulatory elements that enable a plant to sense and respond to a given stress. Our new network analysis approach is focused on the cis-regulatory elements and provides a novel perspective to stress responses. While this analysis method is applied to identifying microgravity responsive regulatory features it can be expanded to identify regulatory elements driving responses to other stresses which can help improve plant growth in extreme conditions on Earth or off-planet.
	Progress continued to be limited because of limited lab access, travel restrictions, and hiring freezes. Although lab work still lags way behind where we should be, we have been moving this forward much better since mid-May. As previously reported, we developed a new method for determining cis-regulatory motif candidates that regulate changes in gene expression in response to altered gravity levels. This method uses machine learning classification to identify motifs that are predictive of the response observed (in this case, is the gene altered in microgravity or not?). Previously we implemented the pipeline and identified candidate genes. In this period, while access to lab was limited, we continued to improve the pipeline in multiple areas. These improvements were benchmarked by comparing the ability to identify known motifs in other stress responses in plants (e.g., cold stress, iron response). We are also working to test the performance in human cell line data where the cis-regulatory elements can be validated by many available ChIP-seq datasets. We benchmark the improvements based on the ability to predict genes in a new dataset that the classifier has never seen. The common criteria is: after the classifier is changed, how good is it at predicting if genes are differentially expressed or not in response to microgravity in a data set the classifier has not seen? One of the major improvements this year (see details below) was in new methods to combine different datasets; the ability to combine datasets had substantial improvement in the predictive score and therefore we are confident in the motifs identified by this classifier. In addition, new RNA-Seq data sets were available in GeneLab allowing us to implement various combinations of training with one or more dataset(s) and testing on an independent one. 1. This year we improved the inputs to the algorithm by resolving nested motifs, co-occurring motifs, and correlated motifs. Two goals for this year that we believe will make this even more useful: a. We want to set up a pipeline to make this easy to port to other species in GeneLab besides Arabidopsis. We have set up a method to scan genomes and identify the motifs of interest. However, running this took over four weeks on the cluster for one plant species (with a relatively small genome), and we need to work on parallelizing this to make it viable

	for easily applying to different plant species.
Task Progress:	2. The initial implementation of the approach only classified motifs that could predict if a gene were identified as differentially expressed in response to microgravity or not. We have employed additional approaches to improve the ability to aggregate across multiple datasets-- already one of the major strengths of this approach over traditional motif finding-- by using clusters of expression as the predictor variable or absolute expression levels. For example, in the original approach, the goal was to identify which motifs could predict if a gene was differentially expressed. To combine experiments, we either classified the gene as differentially expressed in 'any,' 'a majority,' or 'half' of the four GeneLab experiments we were using. The new clustering approach first clusters the genes based on their expression across all four data sets and then asks which motifs can predict the cluster that the gene is classified into. The advantage of this is that it removes the threshold for differential expression, which is subject to the number of replicates in each experiment. The second new approach, predicting the absolute expression level, also avoids the threshold and offers an improvement that it accounts for the basal expression level differences, which is something we know is important based on our cold and iron stress test sets. However, we have yet to figure out how to appropriately score our success on this approach and there are a number of ways to scale across the multiple experiments, so this approach needs more investigation. Three goals for this year that we believe will make this even more useful: a. Using additional GeneLab data that is now available, we would like to incorporate and re-run as is. b. Using pre-defined patterns instead of clusters to focus on genes of interest. For example, if we wanted to look for patterns of things that are low in ground control samples and high in microgravity we could specify this pattern across all replicates. Where most clustering algorithms are data-driven (unsupervised) this supervised approach has been previously implemented in other analysis methods and we can easily compare it to the unsupervised or classification approaches. c. Using a ranking system to validate our expression level score without implementing a threshold cutoff. We have recovered Arabidopsis lines that were stalled or lost over the Covid restrictions and have made plans to complete the yeast one hybrid analysis now that travel is more feasible.
Bibliography Type:	Description: (Last Updated: 05/25/2026)
Abstracts for Journals and Proceedings	Knight M, Nielsen DM, Doherty CJ. "Assessing the Nucleotide-Level Impact of Spaceflight Stress using RNA-Sequencing Data." Southern Section American Society for Plant Biology Annual Meeting, Virtual, April 16-18, 2021. Abstracts. Southern Section American Society for Plant Biology Annual Meeting, Virtual, April 16-18, 2021. , Apr-2021